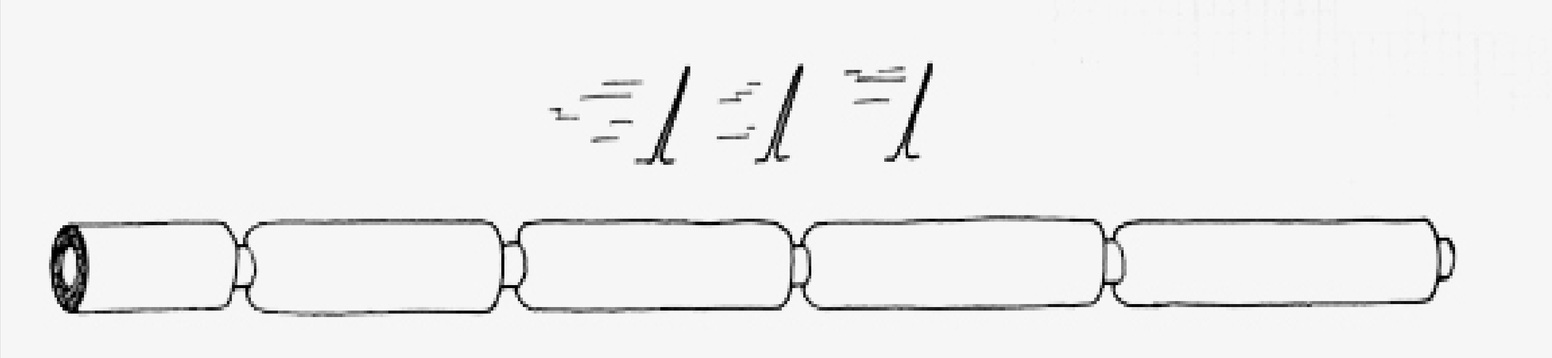
Session Objectives

- 1. With a background of demyelination pathogenesis, discuss and describe the pathological, morphological, histological, and clinicopathological features of **multiple sclerosis**.
- 2. Discuss and describe the pathogenesis, morphological, histological, and clinicopathological features of **Guillian-Barre syndrome.**
- 3. Explain and compare the pathogenesis, morphological, histological, and clinicopathological features of selected **leukodystrophies including metachromatic leukodystrophy, Krabbe's disease, and adrenoleukodystrophy.**
- 4. Discuss the pathogenesis, morphological, histological, and clinicopathological features of **progressive multifocal leukencephalopathy.**

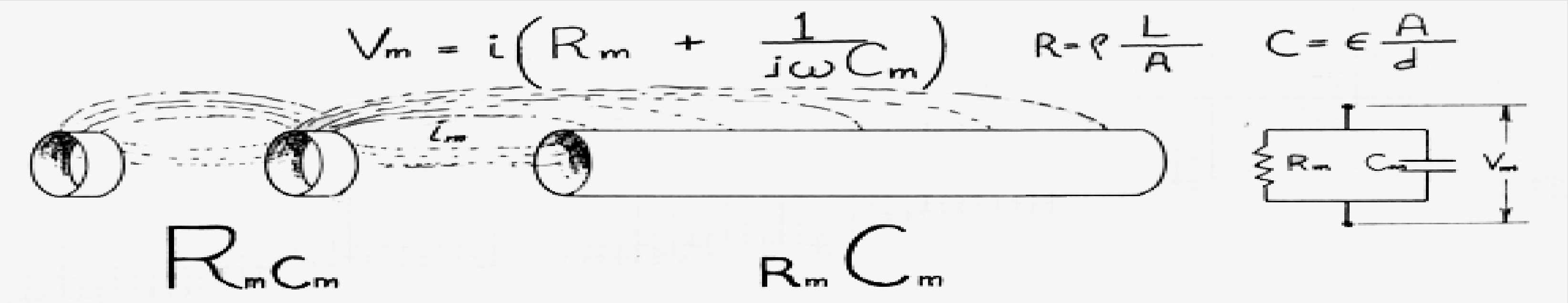
Basics of Demyelination

Effects of demyelination

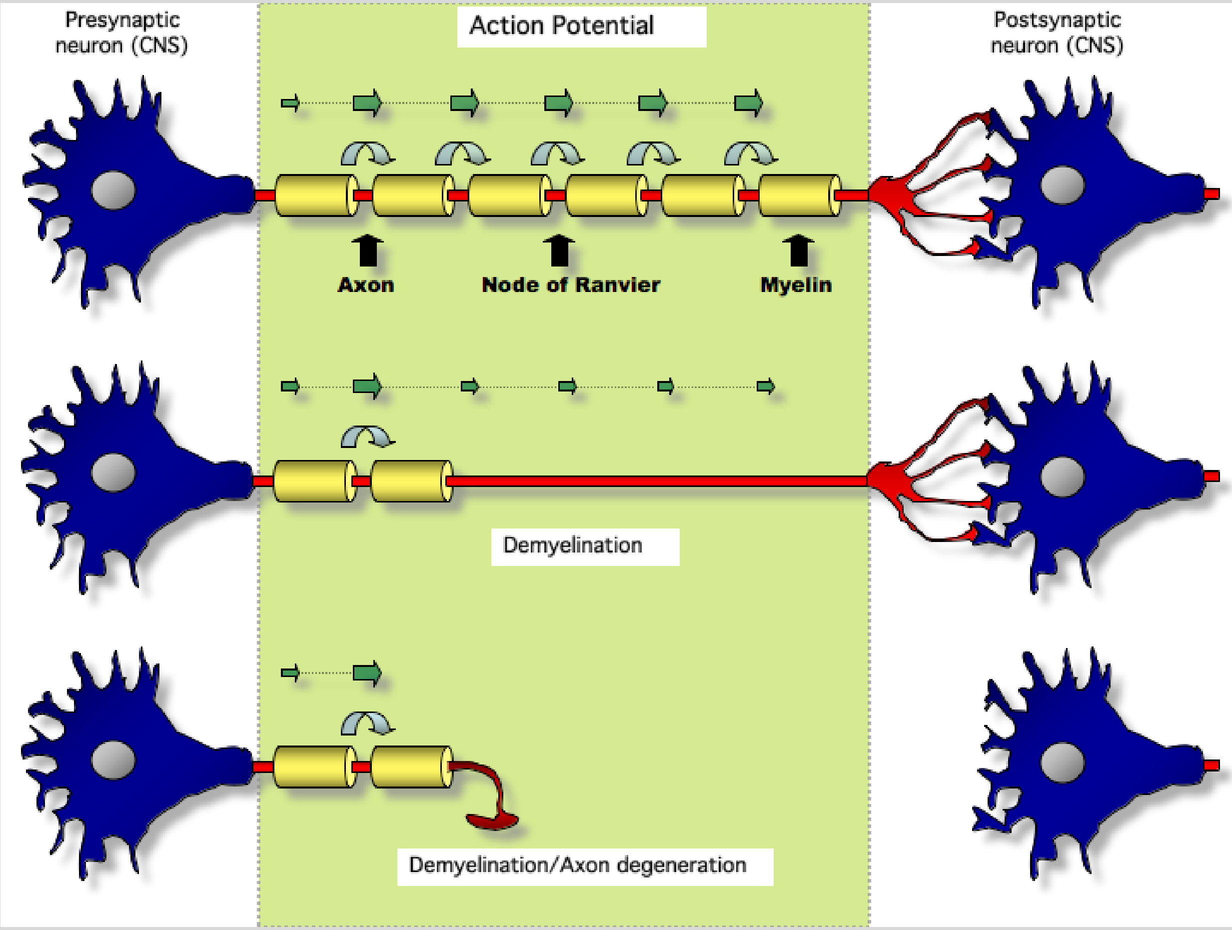
Slowing of conduction velocity due to loss of saltatory conduction.



Conduction block due to impedance mismatch between myelinated and unmyelinated portions of the axon.



NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



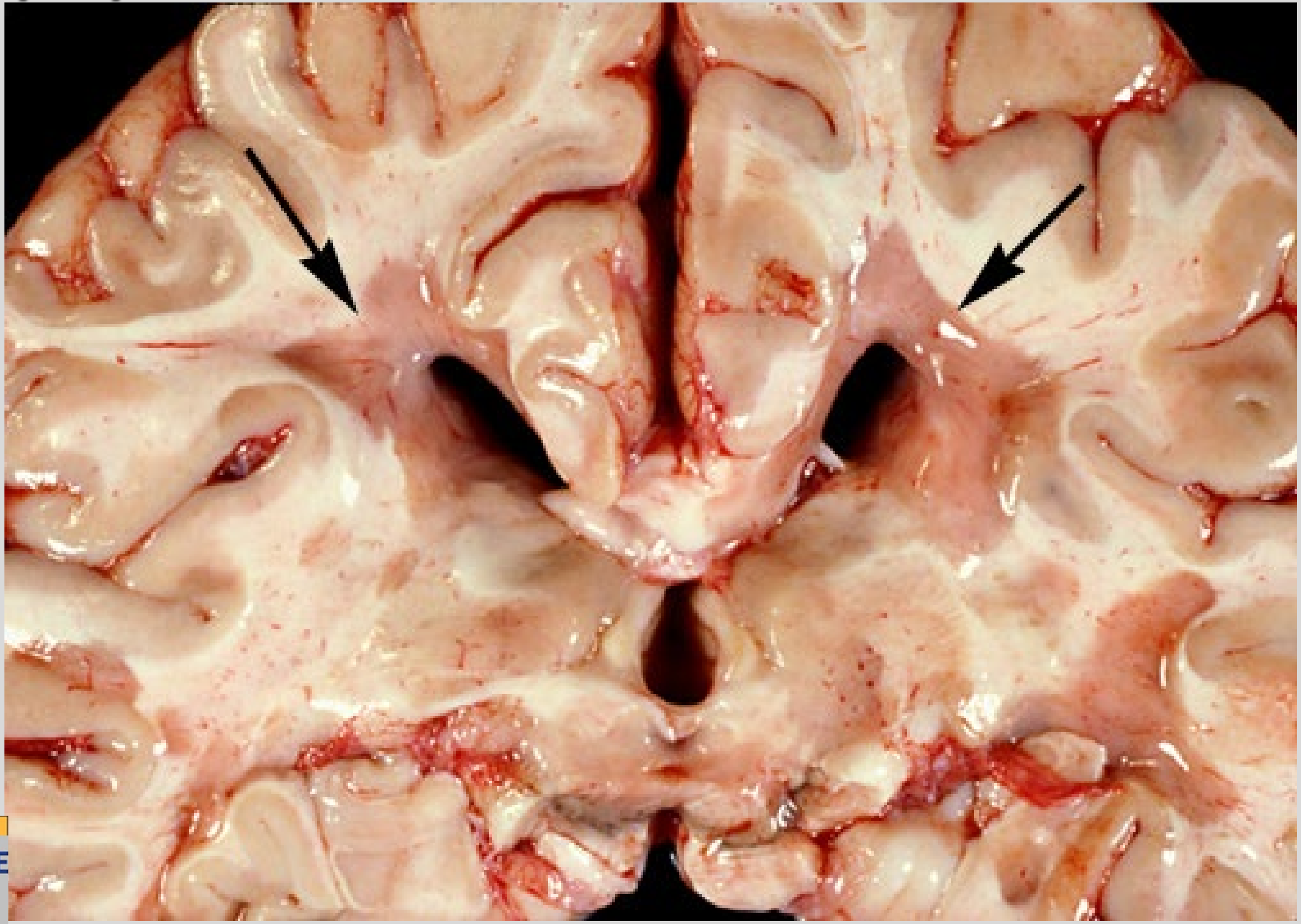
The velocity with which an action potential travels down the axon depends on(a) whether the fiber is myelinated and (b) the diameter of the the fiber. Conduction by local current flow does occur in unmyelinated fibers. However, a faster method of propagation, saltatory conduction, takes place in myelinated fibers.

Case Question

- A 33 year old woman complains of urinary incontinence and blurry vision for the past 3 months. A fundoscopic exam shows no abnormalities. Two months later she develops double vision and numbness in the fingers of her left hand. MRI shows scattered plaques in the patient's brain and spinal cord. Over the next several months, some of these plaques diminish in size, while others appear in new locations. These plaques would most likely show selective loss of which of the following proteins?
- A. beta-amyloid
- B. glial fibrillary acidic protein
- C. synaptophysin
- D. myelin
- E. alpha-synuclein

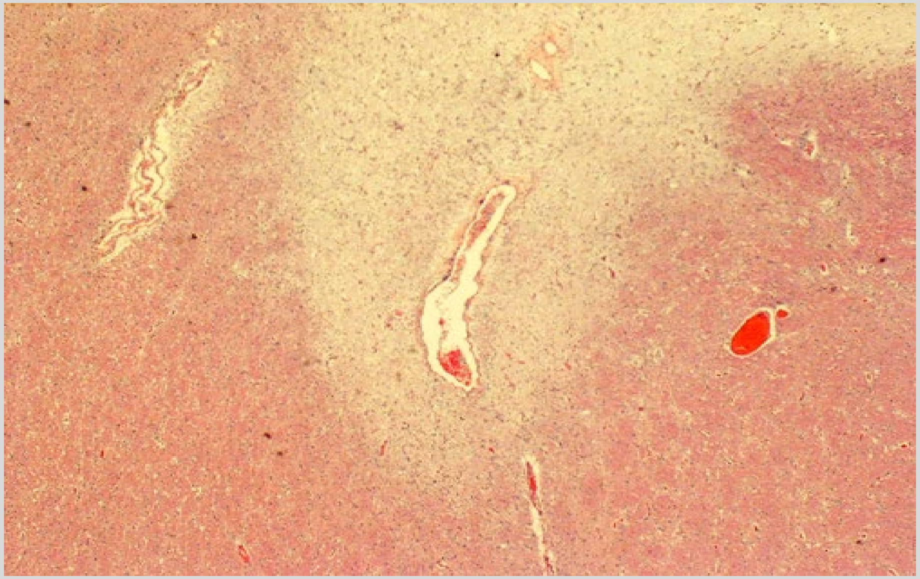
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

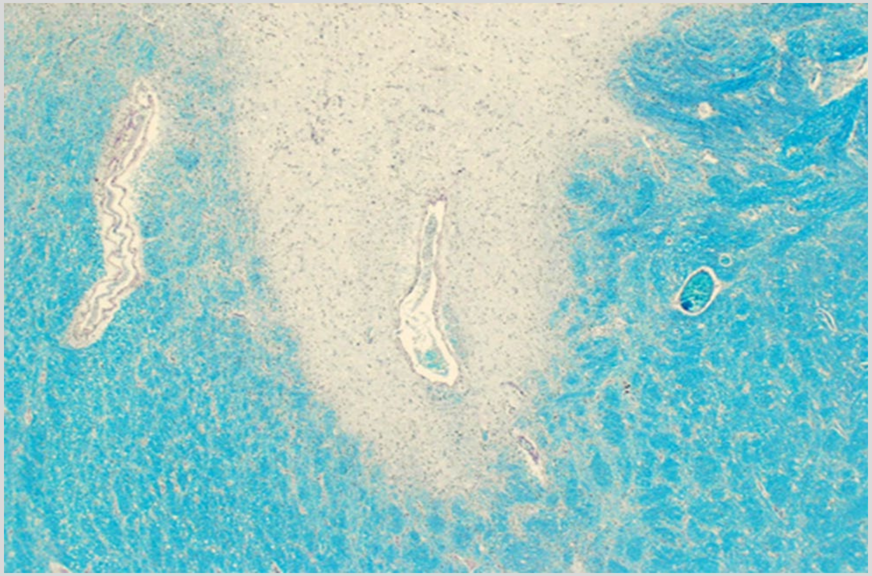
Periventricular, symmetric MS PLAQUES by the lateral ventricles (arrows): **demarcated grey/ grey-pink lesions in unfixed brain tissue; firmer than the adjacent normal white matter (*sclerosis)*. Plaques are often multiple and frequently symmetrical**



https://www.studyblue.com/notes/note/n/cns-pathology-1--2/deck/5478949

- **Periventricular lesion centered around vein.**
- **Older lesion: very little inflammation around the vein.**
- **H and E loss of myelin: lighter pink than surrounding normal white matter (left photo)**





http://missinglink.ucsf.edu/lm/ids_104_Demyelination/Didactic/MSMicroPath.htm

Multiple Sclerosis

- Distinct episodes of neuro deficits, separated in time, attributable to white matter lesions separated in space
- Common *presenting symptoms*:
  - weakness in one or more limbs (40%); spinal cord lesions result in motor and sensory impairment of trunk and limbs, spasticity
  - Optic neuritis (22%) – **Frequent initial manifestation**: unilateral visual impairment due to involvement of optic nerve; (10% to 50% of patients, depending on population) go on to develop sensory disturbances (21%)
  - diplopia (12%)
  - vertigo (8%)
  - bladder dysfunction (5%)
  - Brainstem involvement produces cranial nerve signs (ataxia, nystagmus)
- Symptoms may worsen when ambient temperature is higher than usual (Uhthoff's phenomenon)
- Lhermitte's sign – electrical sensation that runs down back when bending neck

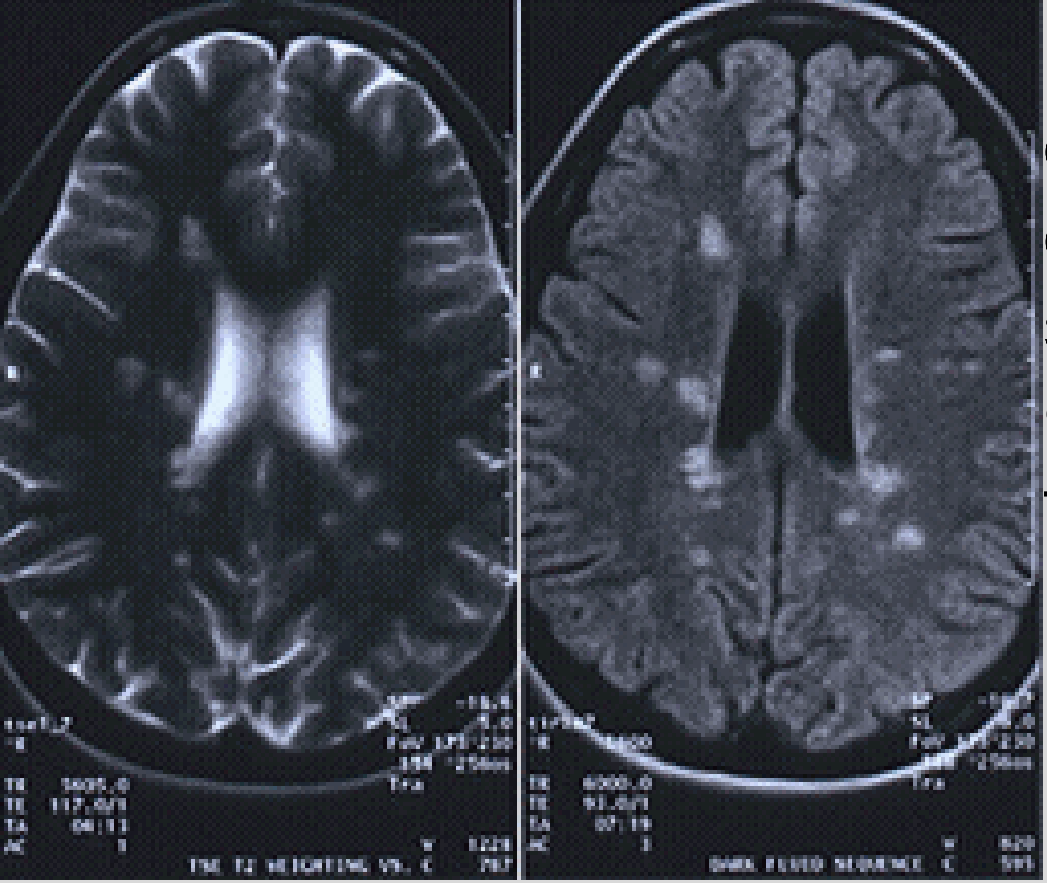
Robbins and Cotran, Pathologic Basis of Disease, 9th edition, 2014, Ch. 28, and 10th edition, 2020, Ch. 28

Multiple Sclerosis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **CSF abnormalities**: mildly elevated protein levels, moderate pleocytosis 1/3 cases (increase mononuclear wbcs), increased IgG levels, and **oligoclonal IgG bands** (immunoelectrophoresis) - presence of a small number of activated B-cell clones in the CNS, thought to be self-reactive
- Evoked response testing may detect slowing in or interruption of conduction in visual, auditory, somatosensory or motor pathways.
- Magnetic resonance imaging – to follow disease progression
- Treatment consists of several types of immunosuppressive or immunomodulatory agents; may slow the progression of the disease but not curative
- Key to diagnosis is recognizing signs and symptoms that would represent multiple lesions, separated by time and in different locations.

Robbins and Cotran, Pathologic Basis of Disease, 9th edition, 2014, Ch. 28, and 10th edition, 2020, Ch. 28



MRI is preferred method of imaging the brain to detect plaques or scarring caused by MS. Often, brains that appear to be normal on CT images will show MS plaques on MRI

The diagnosis of MS cannot be made only on the basis of MR imaging. Other neurologic diseases cause lesions in the brain that appear similar. Normal MR does not absolutely rule out a diagnosis of MS. Approximately 5% of patients who are confirmed to have multiple sclerosis on the basis of other criteria, do not show any plaques or lesions in the brain on MR images.

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Miscellaneous CNS demyelinating diseases

- **Neuromyelitis optica**
  - Syndrome with synchronous (or near-synchronous) bilateral optic neuritis and spinal cord demyelination
  - Once considered a variant of MS, now considered distinct disorder
  - More commonly associated with poor recovery from the first attack
  - Often characterized by the presence of aquaporin-4 antibodies
- **Acute disseminated encephalomyelitis**
  - Diffuse, monophasic demyelinating disease, follows viral infection or, rarely, viral immunization
  - Symptoms usually 1 or 2 weeks later - include headache, lethargy, and coma
  - Rapid clinical course, up to 20% die; remaining patients usually recover completely.
- **Acute necrotizing hemorrhagic encephalomyelitis**
  - Fulminant syndrome of CNS demyelination, usually young adults and children
  - Preceded by recent URI, of unknown cause
  - Fatal in many patients, with significant deficits present in most survivors.

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 28

Case question

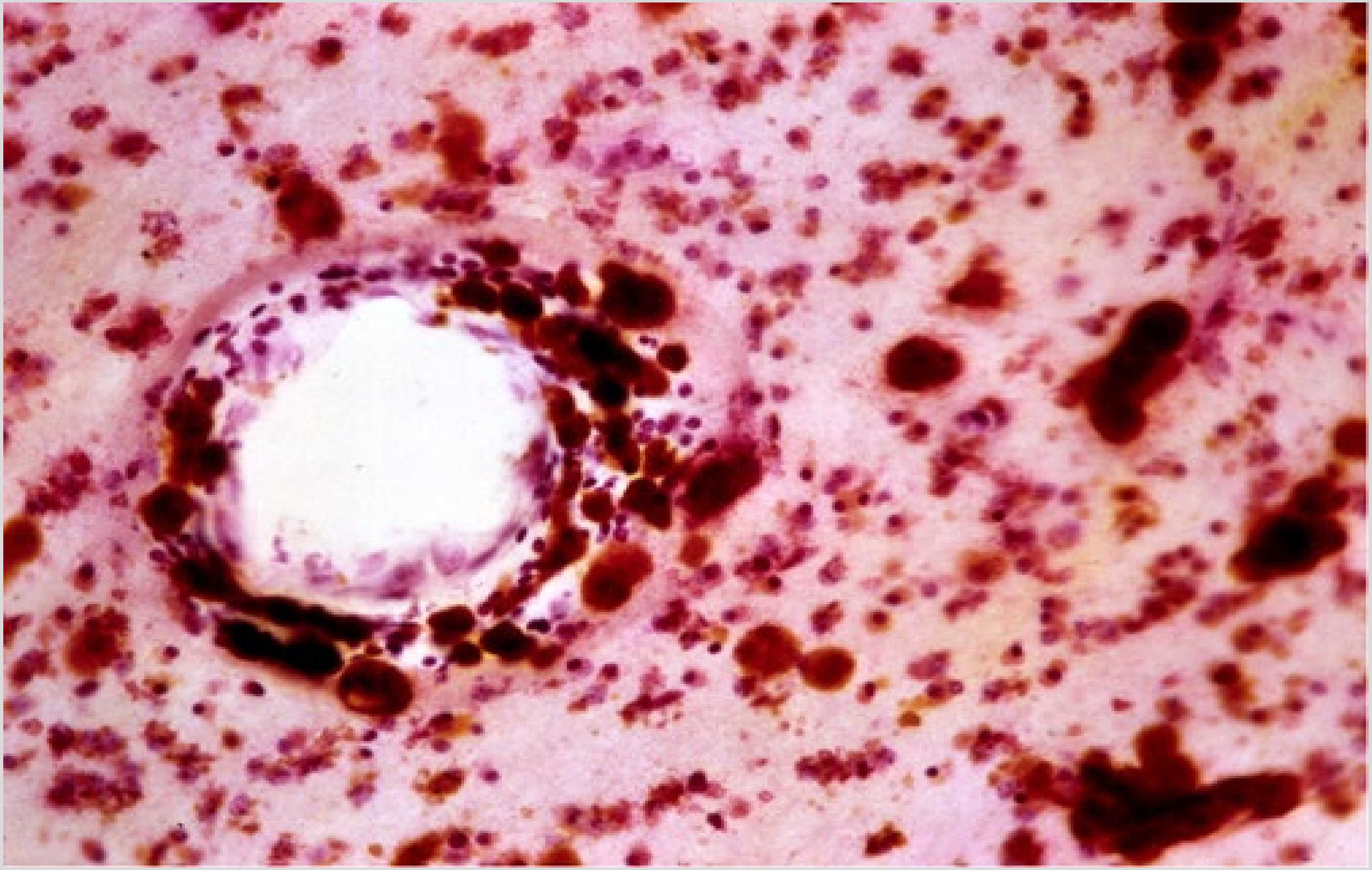
- A 41 year old woman had a flu-like illness for one week. A few days later, she experienced a rapidly progressive, ascending motor weakness that required intubation and mechanical ventilation. A CSF analysis shows clear, colorless spinal fluid under normal pressure with a slightly elevated protein concentration, normal glucose levels, and a cell count with only a few mononuclear cells. The patient recovered in 3 weeks. If lymphocytic infiltrates were seen in peripheral nerves along with segmental demyelination at the initial time she saw the physician, what would be the most likely diagnosis?
- A. Multiple sclerosis
- B. Amyotrophic lateral sclerosis
- C. Varicella-zoster virus infection
- D. Guillain-Barre syndrome
- E. Vitamin B 12 deficiency

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Metachromatic Leukodystrophy

- Autosomal recessive disease - deficiency of lysosomal enzyme, **arylsulfatase A,** which cleaves sulfate from sulfate-containing lipids (sulfatides) as first step in their degradation
- Leads to accumulation of sulfatides, especially cerebroside sulfate, toxic to white matter.
- Histologically - **demyelination with resulting gliosis**
- Macrophages with vacuolated cytoplasm are scattered throughout white matter
- Membrane-bound vacuoles contain complex crystalloid structures made of sulfatides; when they bind to certain dyes (e.g.,toluidine blue) sulfatides shift absorbance spectrum of dye, known as metachromasia
- Similar metachromatic material detected in peripheral nerves and in urine – urine test sensitive method for diagnosis.
- Multiple clinical subtypes

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Ch. 28



Section of brain with metachromatic leukodystrophy stained with cresyl violet. Sulfatides within the macrophages have bound to the stain.

Adrenoleukodystrophy

- **X-linked recessive disease**: loss-of-function mutations in ABCD1 (encodes an ATP-binding cassette transporter protein required for transport of fatty acids into peroxisomes)
- Boys present with behavioral changes and adrenal insufficiency
- Loss of ABCD1 leads to inability to catabolize very-long-chain fatty acids within peroxisomes, resulting in **elevated serum levels of VLCFAs**
- Symptoms due to progressive CNS, peripheral nerve, and adrenal disease
- **CNS myelin loss, gliosis and extensive lymphocytic infiltration**
- **Adrenal cortical atrophy** results in hypocortisolism.

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Ch. 28

Other metabolic genetic diseases

- **Other leukodystrophies** include some with defects in lipid metabolism - have other mechanisms of white matter injury including mutations in genes encoding:
  - Proteins required for myelin formation (Pelizaeus-Merzbacher disease)
  - Intermediate filament protein GFAP (Alexander disease)
  - Translation initiation factor eIF2B (vanishing white matter leukoencephalopathy) .
- **Neuronal Storage Diseases** – large class of diseases with genetic heterogeneity usually caused by enzyme deficits or defects in protein or lipid trafficking within neurons -
  - result in accumulation of storage material within neurons, followed by neuronal death
  - leads to seizures and generalized loss of neurologic function
- **Mitochondrial Encephalomyopathies** - Disorders of energy generation causing a range of neurologic disease, often associated with abnormalities in other tissues. May include:
  - Mitochondrial encephalomyopathy, lactic acidosis, and strokelike episodes **(MELAS)** - recurrent episodes of acute neurologic dysfunction, cognitive changes, muscle weakness, and lactic acidosis
  - Myoclonic epilepsy with ragged red fibers (MERRF) resulting inmyoclonus (a seizure disorder) and a myopathy showing ragged red fibers on muscle biopsy
  - Leigh syndrome – Infants have lactic acidemia, arrest of psychomotor development, feeding problems, seizures, extraocular palsies, and weakness with hypotonia; death usually occurs within 1 to 2 years

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Ch. 28

Case Question

- A 6 month old girl with metachromatic leukodystrophy fails to thrive and dies of respiratory insufficiency. Histological examination of the child's brain at autopsy shows marked accumulation of metachromatic material in the white matter and prominent astrogliosis. Lab studies confirm an accumulation of cerbroside. This patient most likely suffered from an inborn error of metabolism, characterized by a deficiency in which of the following enzymes?
- A. galactosidase
- B. mannosidase
- C. N-acetyl-hexosaminidase
- D. arylsulfatase A
- E. phenylalanine hydroxylase

Case question

- An 8 month old boy exhibits severe motor, sensory, and cognitive impairments. Brain biopsy shows a disease of white matter characterized by the accumulation of "globoid cells". Biochemical studies reveal an absence of galactocerbroside beta-galactosidase activity. What is the appropriate diagnosis?

- A. Alexander disease
- B. Hurler disease
- C. Krabbe disease
- D. Metachromatic leukodystrophy
- E. Adrenoleukodystrophy

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Case question

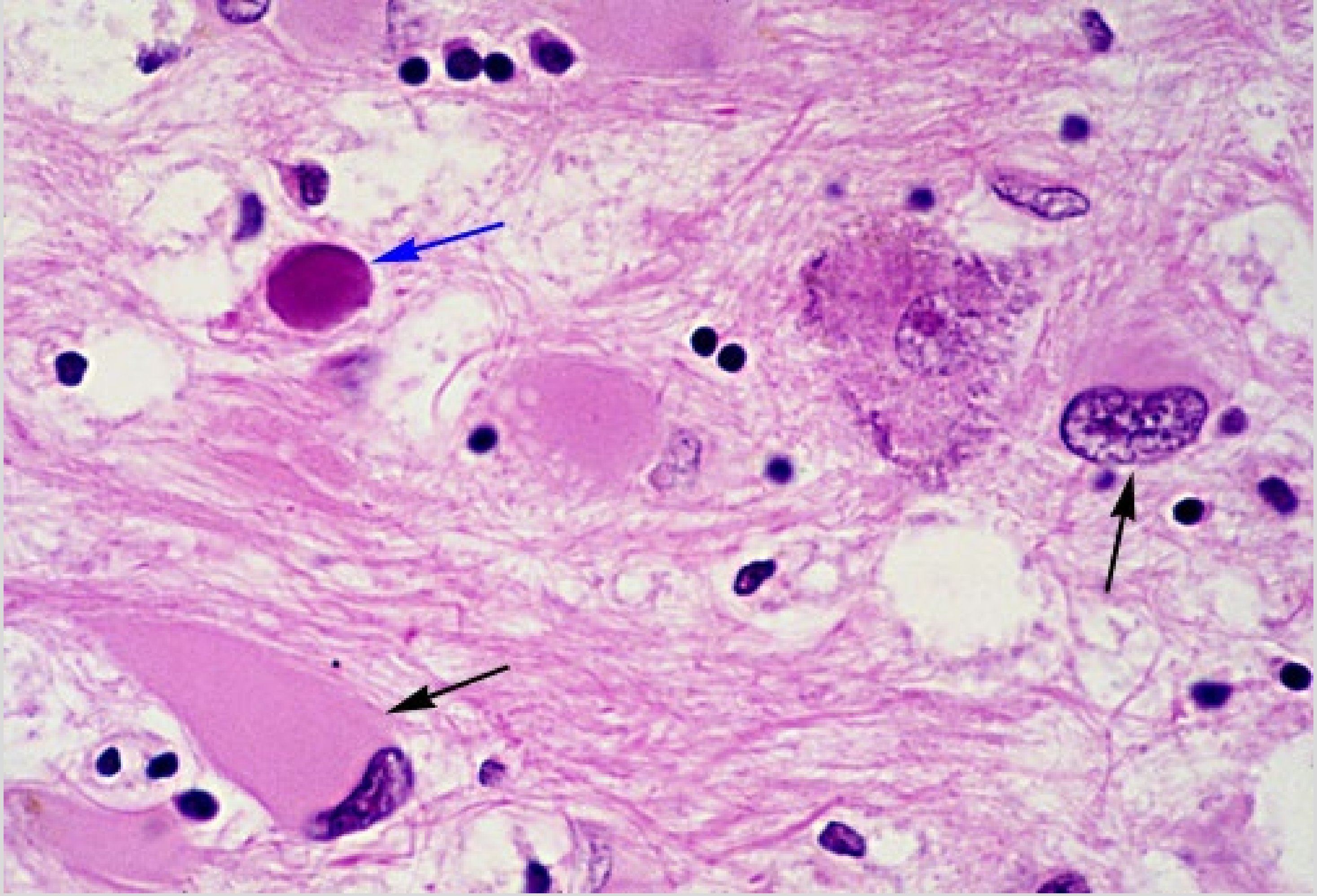
- A 27 year old man presents with dementia, weakness, visual loss, and ataxia one year after receiving a cadaveric kidney transplant. Which of the following is the most likely cause of this patient’s CNS disorder?
- A. Adrenoleukodystrophy
- B. Gaucher disease
- C. Metachromatic leukodystrophy
- D. Progressive multifocal leukoenceophalopathy
- E. Subacute sclerosing panencephalitis

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Progressive Multifocal Leukoencephalopathy

- Immunosuppressed patients, including transplant patients and AIDS patients, are at risk. Immunocompetent people usually do not get PML
- Clinical – focal (depending on location),progressive neurologic symptoms and signs
- Extensive, usually multifocal, cerebral or cerebellar demyelinative lesions can be seen on CT and MRI

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Ch. 28



PML (H&E): reactive astrocytes (black arrows), frequently pleomorphic .

Blue arrow = oligodendrocyte nucleus markedly enlarged with viral particles, giving it a magenta color.

Summary Slide

- Review of basic demyelination
- Multiple Sclerosis – chronic, autoimmune, demyelinating disease
  - Environmental and genetic
  - Plaques separated in space and time, inflammation and demyelination
  - Both motor and sensory disturbances
  - CSF – oligoclonal bands
- Guillain-Barre syndrome – acute, immune, inflammation and segmental demyelination, may be preceding viral illness, ascending paralysis

Summary Slide (continued)

- Leukodystrophies
  - Metachromatic leukodystrophy – deficiency in arylsulfatase-A results in sulfatide accumulation which causes myelin breakdown, AR, sulfatides bind stain and changes color
  - Krabbe's disease – deficiency in galactocerebroside beta-galactosidase which shunts to production of galactosylsphingosine, toxic to oligodendrocytes – "globoid cells" around blood vessels
  - Adrenoleukodystrophy – X- linked, lack of catabolism of VLCFAs, atrophy of adrenal gland, behavioural changes

- PML – JC polyomavirus infects oligodendrocytes which causes demyelination, immunocompromised patients